Overview of Logistic Regression

- **Example for Logistic Regression**

Learning Objectives

- Perform logistic regression on a
  real-world dataset using statistical software

- Analyze the simplest logistic regression model by hand

Example: Coronary Heart Disease

The coronary heart disease dataset.

- Response: Y (CHD): **0 = No disease**, **1 = Disease**
- Independent variable: X (**Age**)
- There are 100 subjects: 43 with disease and 57 no-disease.

Questions: Is Age associated with CHD? How to predict CHD from Age?

R code: suppose that we save the data file in the folder “C:/Temp”

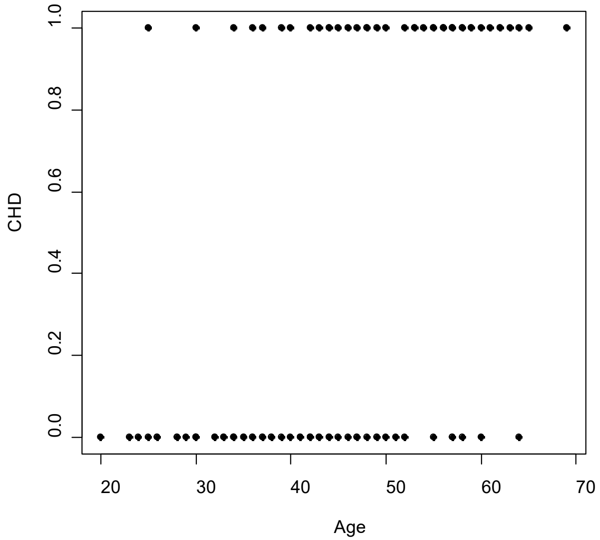
- **data0 < read.table(“C:/Temp/chdage.csv", head=T, sep=",")**

Data

```
> data0[1:5,]
```

	ID	Age	CHD
1	1	20	0
2	2	23	0
3	3	24	0
4	5	25	1
5	4	25	0

```
> attach(data0)
> plot(Age, CHD)
```

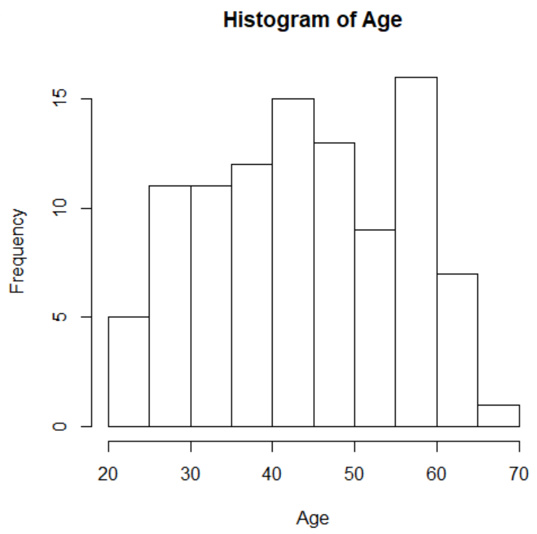


Empirical Data Analysis

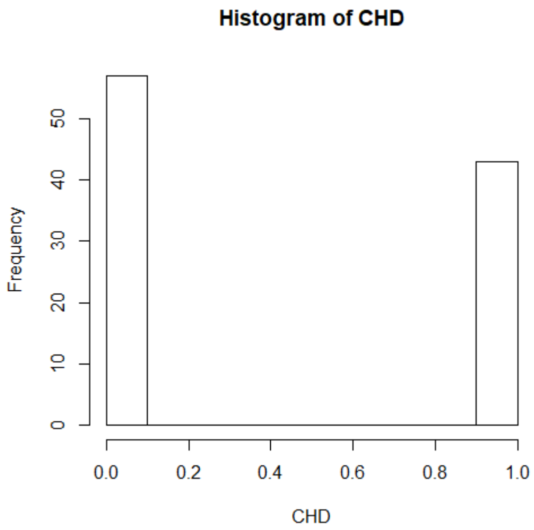
```
> summary(data0)
       ID              Age              CHD
 Min.   :  1.00    Min.   :20.00    Min.   :0.00
 1st Qu.: 25.75    1st Qu.:34.75    1st Qu.:0.00
 Median : 50.50    Median :44.00    Median :0.00
 Mean   : 50.50    Mean   :44.38    Mean   :0.43
 3rd Qu.: 75.25    3rd Qu.:55.00    3rd Qu.:1.00
 Max.   :100.00    Max.   :69.00    Max.   :1.00
## In general we should put R output in the appendix of your report!
```

Histogram Plots

> hist(Age)



>hist(CHD)



Logistic Regression Model

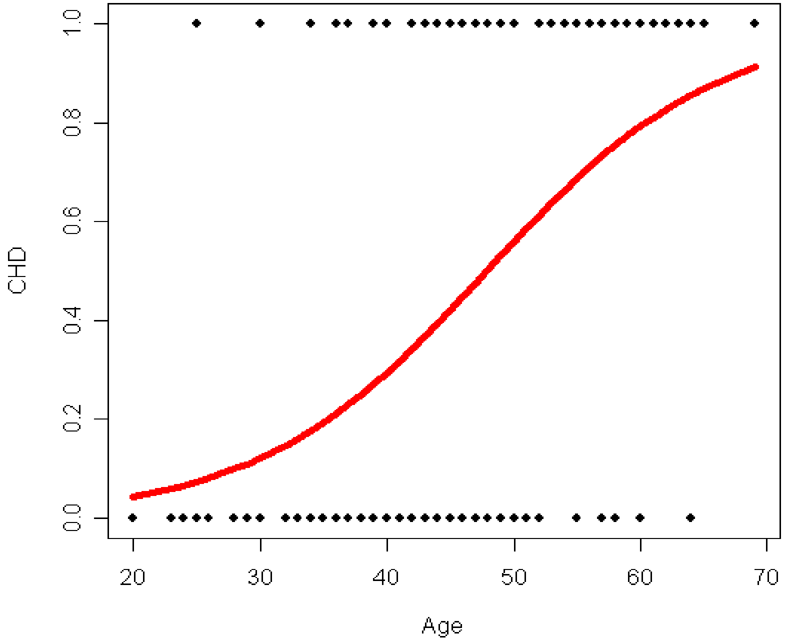
```
>glm1 <- glm(CHD ~ Age, family=binomial(link="logit"), data=data0)
>summary(glm1)
```

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-5.30945	1.13365	-4.683	2.82e-06 ***
Age	0.11092	0.02406	4.610	4.02e-06 ***

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = -5.3095 + 0.1109\ \mathrm{Age}$$

Logistic Regression

```
plot(Age, CHD)
lines(Age,fitted.values(glm1),
col="red")
```



CI of coefficients in Logistic

- How to find the $\mathbf{1-\alpha}$ Confidence Interval on $\boldsymbol{\beta_0}, \boldsymbol{\beta_1}$ in R ?

$$\text{logit}(\pi_i) = -5.3095 + 0.1109\ \text{Age}_i$$

> confint(glm1, level=0.95)

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	-7.72587162	-3.2461547
Age	0.06693158	0.1620067

Simplest Logistic Regression Model

In the CHD dataset, define a new variable $\mathrm{Flag} = I(\mathrm{Age} \geq 50)$.

Objective: Will $\mathrm{I}(\mathrm{Age} \geq 50)$ be a risk factor of CHD?

Model: the simplest logistic regression model

$$\log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_1 \, \mathrm{Flag_i}$$

Statistical Questions:

- Test $H_0 \colon \beta_1 = 0$ against $H_1 \colon \beta_1 \neq 0$

Analysis by R

```
> flag <- I(Age >=50);
> glm2 <- glm(CHD ~ flag, family = binomial(link="logit"), data = data0);
```

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	**-1.0380**	**0.2822**	-3.678	0.000235 ***
flagTRUE	**2.0989**	**0.4788**	4.384	**1.17e-05 ***

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \mathbf{-1.0380} + \mathbf{2.0989}\, I(\text{Age} \geq 50)$$

Analysis by Hand

	CHD+ (Y=1)	CHD– (Y=0)	Sum
Age >= 50 (X=1)	$a = 26$	$b = 9$	35
Age < 50 (X=0)	$c = 17$	$d = 48$	65

When fitting $\log\frac{\pi_i}{1-\pi_i} = \beta_0 + \beta_1 \text{ Flag}_\text{i}$, we have

$$\widehat{\boldsymbol{\beta}}_0 = \mathbf{log}\left(\frac{c}{d}\right) = \mathbf{log_e}\left(\frac{17}{48}\right) = -1.0380,\ \ \widehat{\boldsymbol{\beta}}_1 = \mathbf{log}\left(\frac{ad}{bc}\right) = \mathbf{log_e}\left(\frac{26\times 48}{9\times 17}\right) = 2.0989$$

The standard error

$$\mathbf{se}(\widehat{\boldsymbol{\beta}}_0) = \sqrt{\frac{1}{17}+\frac{1}{48}} = 0.2822, \qquad \mathbf{se}(\widehat{\boldsymbol{\beta}}_1) = \sqrt{\frac{1}{26}+\frac{1}{9}+\frac{1}{17}+\frac{1}{48}} = 0.4788$$

Testing Hypothesis

In the problem of testing $H_0: \beta_1 = 0$ against $H_1: \beta_1 \neq 0$

- The testing statistic is given by

$$Z_{obs} = \frac{\boldsymbol{\hat{\beta}_1} - 0}{\mathbf{se}(\boldsymbol{\hat{\beta}_1})} = \frac{2.0989 - 0}{0.4788} = 4.3837$$

- The p-value is $\boldsymbol{P}(|Z| > |Z_{obs}|)$

$$= 2\boldsymbol{P}(N(0,1) > 4.3837) = \mathbf{1.166804}\boldsymbol{e} - \mathbf{05}$$

(**R code**: 2 * (1-pnorm(4.3837))

- Thus we conclude that Age ≥ 50 is a risk factor of CHD.

Summary

- Logistic regression in R (**glm**)
- Plots for logistic regression
- Simplest logistic regression by hand